Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	19 Feb 2026
Team ID	LTVIP2026TMIDS83753
Project Name	Weather-Based Prediction of Wind Turbine Energy Output: A Next-Generation Approach to Renewable Energy Management

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Landing Page	System shall display a landing page. System shall provide "Get Started" button. System shall navigate to dashboard.
FR-2	Weather Retrieval	System shall allow user to select city. System shall fetch weather information. System shall display weather details.
FR-3	Energy Prediction Input	System shall allow user to enter: • Wind Speed (m/s) • Theoretical Power (KWh) System shall validate input fields.
FR-4	Prediction Processing	System shall send input data to Flask backend. System shall process input using Random Forest model. System shall generate predicted energy output.
FR-5	Result Display	System shall display predicted energy output. System shall allow user to modify input and predict again.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Interface shall be simple and user-friendly. Dashboard shall be responsive
NFR-2	**Security**	System shall validate input to prevent errors.
NFR-3	**Reliability**	System shall handle invalid inputs. System shall not crash for incorrect values.
NFR-4	**Performance**	Prediction shall be generated within 2–3 seconds.